

# SHOU-JEN (LAWRANCE) CHEN

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## Summary

- Machine learning engineer with hands-on experience building LLM-powered content generation pipelines, multimodal AI systems, and agentic workflows that move from research to production. Skilled in fine-tuning techniques including DPO and RLHF, and rigorous evaluation design — with a documented habit of going deep through multi-agent research, published contributions, and client-facing deployments. Strong in Python and PyTorch, and most energized by the high-ownership, 0→1 problems that define early-stage AI products.

## EDUCATION

**Electrical Engineering & Computer Science, University of California, Berkeley (UCB)** **Berkeley, California**  
*Master of Engineering, Area of Concentration: Robotics and Embedded Software* *Expected Date: Oct 2025 - May 2026*  
• Semester 1 GPA (ongoing): **4.0/4.0** (Selected Courses: Machine Learning, Embedded Systems, Deep RL, Data Science)

**Electrical Engineering, National Taiwan University (NTU)** **Taipei, Taiwan**  
*Bachelor of Science in Engineering* *Sep 2020 - Jun 2025*  
• Dean's List: 2023 Fall (**Top 5%**) & **NTU Admissions Scholarship** for Overseas Chinese Students

**Core Skills:** Python, Machine Learning, Data Structures & Algorithms, PyTorch, Reinforcement Learning, C++, MuJoCo, ROS

## PROFESSIONAL EXPERIENCE & PROJECTS

**Software Engineer** **Berkeley, California**  
*JPMorganChase Software Engineering Job Simulation on Forage* 🏢 **JPMorganChase** *April 2026*

- Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework.
- Implemented transaction validation and persistence logic with Spring Data JPA and an H2 SQL database, including entity modeling and balance updates across relational User records.
- Connected the service to an external REST Incentive API using RestTemplate and exposed a balance-query endpoint via a Spring controller, returning JSON responses while maintaining clean architectural boundaries.
- Verified system behavior using Maven test suites and debugger-driven inspection, ensuring reliability across message ingestion, database operations, and external API interactions.

**Reinforcement Learning Researcher** **Berkeley, California**  
*Risk-Triggered Latent-Space Corrections for Near-Failure Recovery* 🏢 **UC Berkeley** *Feb 2026 – May 2026*

- Designed a concurrent decoupled correction architecture for Soft Actor-Critic (SAC) where a lightweight MLP applies additive latent corrections  $\Delta z$  to the actor's hidden representation; a geometric near-failure trigger (sliding-window range check on hand-to-object and object-to-goal distances) gates corrections to only **2–12%** of steps at convergence.
- Identified and resolved co-adaptation via gradient isolation: applied stop-gradient on actor encoder during correction update, preventing encoder from learning latents and ensuring base actor remains independently capable at deployment.
- Evaluated across 4 Meta-World manipulation tasks; improved mean final success rate by **+57pp** on push-v3 (32.5%→89.5%) and **+27.5pp** on pick-place-v3 (66.0%→93.5%), achieving full seed reliability (4/4) versus 0/4 under naive concurrent training.

**Machine Learning Engineer / AI Engineer (Capstone Project)** **Berkeley, California**  
*AI-driven Personalized Assisted Device, UC Berkeley* *Aug 2025 – Ongoing*

- Built a 3-phase LLM-guided optimization pipeline over a 15-dimensional design space, converging to a personalized assistive device design in as few as 2 iterations and achieving **83% overall convergence rate** across 12 runs and 4 participants.
- Designed task interfaces across 8 simulated grasping environments, enabling evaluation of grip force and tasks success rates.
- Optimized designs reduced required grip force to a mean of 2.5 Nm, with **7 of 8 tasks achievable at or below 1.7 Nm**.

**Machine Learning Engineer / Robotics Engineer** **Berkeley, California**  
*IMU-Controlled Quadruped Robot* 🏢 **UC Berkeley** *Aug 2025 – Dec 2025*

- Designed real-time gait generation and inverse kinematics for an 8-DoF quadruped, enabling walking, turning, and jumping.
- Integrated Raspberry Pi control with Dynamixel actuators over UART, achieving synchronized motor commands at up to 100 Hz.
- Prototyped a wearable EMG-IMU interface and evaluated machine-learning-based gesture recognition for robot control.

**Machine Learning Engineer** **Berkeley, California**  
*Vision Fairness Analysis — CNN vs Vision Transformer* 🏢 **UC Berkeley** *Aug 2025 – Dec 2025*

- Implemented PyTorch training and evaluation pipelines comparing CNN and transformer architectures on diverse facial datasets.
- Evaluated image reconstruction and classification performance using PSNR, SSIM, and accuracy across multiple test conditions.
- Analyzed performance consistency across demographic groups under controlled distribution shift and training imbalance

## Industry/Lab Researcher

Taipei, Taiwan

Industry-University Collaboration, **TrendForce Corporation**

Oct 2024 – Jun 2025

- Developed automated content generation service integrating LLM APIs with structured prompt templates.
- Built LLM-powered text generation workflow with controllable style parameters and automated quality evaluation loops.
- Iteratively refined prompts and model configurations to improve consistency and reliability of generated outputs.
- Implemented style-controlled generation via parameterized prompts and evaluation loops for TrendForce newsletter.

## Intern: Software Engineer Team

Taipei, Taiwan

Software Engineer Internship, **Pegatron Corporation**

Jul 2023 – Aug 2023

- Collaborated in a team of three to develop “Visual Prompt Defect Detection” for an internal internship project competition, receiving **1st Place** among four total groups. Our project focuses on using in-context learning.
- Designed UI using Gradio for project demo & built backend data ingestion pipeline storing image datasets to storage via APIs.
- Completed training on **Git, Docker, Kubernetes, Python, Pandas, Numpy, and MongoDB**.

## Robotics Engineer

Taipei, Taiwan

Embedded System Lab Project  **NTU**

Aug 2023 – Jan 2023

- Collaborated with a teammate to develop a smart driving system, featuring real-time notifications and auto-parking.
- Integrated PiRacer Pro and STM32 board to boost system capabilities with analytics and machine learning.

## Intern: Software Engineer and Webpage Developer

Taipei, Taiwan

Software Engineer Internship, **IADY Photonics**

Jul 2020 – Aug 2020 / Jul 2021 – Aug 2021

- Designed software enabled auto-driving and basic sensor mapping using Robot Operating System on Linux.
- Assembled robotic devices, including constructing the hardware framework and configuring wires, running on Raspberry Pi.
- Initiated a Kickstarter Campaign in a team of 5 for the product, Genix Lite and designed company website using HTML and CSS.

## PUBLICATIONS & CONFERENCE EXPERIENCE

- **Co-First Author, COLM 2024**, “LLM Discussion: Enhancing the Creativity of LLMs via Discussion Framework and Role-Play” 
- **Co Author, ICLR 2025**, “Dynamic-SUPERB Phase-2: An Open Benchmark Evolving through Collaborative Expansion for Comprehensive Evaluation of Spoken Language Models with 180+ Tasks” 
- **Reviewer, 2024 IEEE Spoken Language Technology Workshop (SLT)**, Reviewed and evaluated research paper.

## RESEARCH EXPERIENCE

### Robot Learning Laboratory (RLL), NTU

Taipei, Taiwan

Research Assistant / Undergraduate Researcher | Advisor: **Prof. Shao-Hua Sun**

Oct 2023 – Jun 2025

- Introduced LLM Discussion framework that enhances the creativity of large language models via multi-agent discussion and role-playing, surpassing single-agent and existing multi-agent LLM frameworks in creativity tests by 20%.
- Designed an LLM evaluation focused on creativity metrics, achieving a 0.7 correlation with human on 1,400+ responses.
- Develop a decision-making agent that optimizes selection between in-context learning with Retrieval-Augmented Generation, supervised fine-tuning, and reinforcement learning from human feedback in a streamline learning process.

### Speech Processing and Machine Learning Laboratory (SPMLLab), NTU

Taipei, Taiwan

Undergraduate Researcher | Advisor: **Prof. Hung-yi Lee**

Jul 2023 – Jun 2025

- Devised and preprocessed a “SoundEffectDetection” dataset/task for Dynamic-SUPERB, aimed at building universal speech models capable of leveraging instruction tuning to perform multiple tasks in a zero-shot fashion. 
- Utilize Direct Preference Optimization (DPO) reinforcement learning as a method for Reinforcement Learning from Human Feedback (RLHF) to train a text-instruction-guided voice conversion model, aiming to enhance model performance.

### Wireless and Mobile Networking Laboratory (WMNLab), NTU

Taipei, Taiwan

Undergraduate Researcher | Advisor: **Prof. Hung-Yu Wei**

Jan. 2023 – Jan. 2024

- Conducted mobile phone experiments on the MRT using user datagram protocol to evaluate network handover.
- Analyzed data and presented findings on experiments across a 25-km route to professors and graduate students.

## OTHER EXPERIENCE

### Generative Artificial Intelligence Course, NTU (Technical Report / Course Webpage

Taipei, Taiwan

Teaching Assistant (TA)

Feb. 2024 – Jun. 2024

- Collaborated with 30 TAs to manage and respond to student emails for a class of over 1,000 students and 1,000 auditors.
- Designed and graded a coursework related to calling API alongside two other TAs and provided 8 office hours.

### English Written and Oral Communication for Professional Pursuit Course, NTU

Taipei, Taiwan

Teaching Assistant (TA)

Expected Date: Sept. 2024 – Jan. 2025

- Dedicate over 30 hours per month throughout the semester to support course instruction and weekly reminders.
- Manage and promptly respond to emails and inquiries for a class of over 70 students, ensuring timely communication.